

Jesús Vega Ferrero

Astrophysicist · Data Scientist

A. Personal Information

e-mail: astrovega@gmail.com

Orcid ID: [0000-0003-2338-5567](https://orcid.org/0000-0003-2338-5567)

A.1. Current position

Nov 2025 Lead Data Scientist
Present Deduce Data Solutions S.L.
Torres-Quevedo fellow

A.2. Previous positions

Feb 2024 Tenure Track researcher
Oct 2025 Centro de Estudios de Física del Cosmos de Aragón (CEFCA)
Spatially resolved properties of galaxies and low-surface brightness structures

Feb 2023 Postdoctoral researcher at Universidad de Valladolid (UVa)
Jan 2024 Detection of galactic stellar disk truncations for the Euclid mission (ESA)
Supervisor: **Prof. F. Buitrago Alonso**

Sep 2021 Postdoctoral researcher at Instituto de Astrofísica de Canarias (IAC)
Jan 2023 Galaxy evolution with artificial intelligence
Supervisor: **Dr. M. Huertas-Company**

Jul 2020 Postdoctoral researcher at Instituto de Física de Cantabria (IFCA-CSIC)
Aug 2021 and Instituto de Investigación de Inteligencia Artificial (IIIA-CSIC)
Machine Learning applications to the detection and redshift estimation of X-IFU/Athena sources
Supervisors: **Dr. M. T. Ceballos Merino** and **Dr. F. J. Carrera Troyano**

Jan 2019 Postdoctoral researcher at University of Pennsylvania
Dec 2019 Deep-learning galaxy morphologies for the Dark Energy Survey (DES)
Supervisor: **Prof. M. Bernardi**

Jan 2018 Postdoctoral researcher at Instituto de Física de Cantabria (IFCA-CSIC)
Dec 2018 Visiting scholar at University of Pennsylvania
Dark matter distribution in galaxy clusters
Supervisors: **Dr. J. M. Diego Rodríguez** and **Prof. G. Bernstein**

Aug 2016 Postdoctoral researcher at University of Pennsylvania
Aug 2017 Free-form models for the six Hubble Frontier Fields Clusters
Supervisors: **Prof. G. Bernstein** and **Dr. J. M. Diego Rodríguez**

A.3. Education

Feb 2011 PhD degree in Astrophysics with a distinction "cum laude"

Jul 2015 Thesis entitled: *"Dark matter distribution in the Universe with gravitational lensing: the distribution function of Einstein radii as cosmological probes"*

Supervisors: **Prof. Gustavo Yepes** and **Prof. David Valls-Gabaud**

Universidad Autónoma de Madrid (Spain) and Observatoire de Paris (France)

2010 Master degree in Astrophysics

Thesis entitled: "¿Cuántos anillos de Einstein hay en el Universo?"

Supervisor: **Prof. David Valls-Gabaud**

Universidad Complutense de Madrid and Universidad Autónoma de Madrid (Spain)

2009 Degree in Physics, specialized in **Astrophysics**

Universidad Complutense de Madrid (Spain)

Università degli Studi di Palermo (Italy)

Schools

2012 XXIV Canary Islands Winter School of Astrophysics:

Astrophysical applications of gravitational lensing (Tenerife, Spain)

2012 21st Summer School of Parallel Computing (Bologna, Italy)

2007 1^a Escuela Avanzada ASTROCAM de Astrofísica y Cosmología: "Getting ready for upcoming IR to mm observing facilities" (Madrid, Spain)

A.4. Technical skills and training

- **Scientific programming and GPU computing:** *Python, PyTorch, TensorFlow*; experience training and applying ML models on GPU resources.
- **Astronomical image processing and source detection:** [GNUastro](#)/[NoiseChisel](#), [AstroPhot](#), [StarNet](#); PSF modeling/subtraction, photometry and low-surface-brightness imaging.
- **Machine learning and computer vision:** supervised, self-supervised and unsupervised learning; CNNs, foundation models, segmentation and classification.
- **Large astronomical datasets:** [IllustrisTNG project](#), [Dark Energy Survey](#) and [MultiDark, Euclid, J-PLUS/J-PAS](#).
- **Astrophysics:** Galaxy formation and evolution, Galaxy Clusters, Gravitational Lensing, Cosmology, and N-body Cosmological Simulations.

A.5. Languages

Spanish (mother tongue), **English** (proficient user) and **Italian** (proficient user)

B. CV Summary

General Quality Indicators (ADS)

Publications in Q1 journals: 34; h-index: 23

Total number of citations: 3747; Average citations: 110.2

Contributions in conferences: 26 (2 invited talks, 4 seminars, and 20 oral presentations)

My research combines **observational astrophysics, astronomical image processing and artificial intelligence**, with particular experience in extracting and classifying faint or complex structures from large imaging datasets. Over the last decade, I have worked across galaxy evolution, gravitational lensing and low-surface-brightness science, increasingly developing and applying ML and computer-vision methods to astronomical problems.

Currently, as **Lead Data Scientist** at DEDUCE DATA SOLUTIONS S.L., I focus on developing **innovative AI-based solutions across a range of industrial sectors**. During my first year in the private sector, I have **taken on increasing technical responsibility across several R&D projects**, contributing from problem definition and methodological design through data analysis, model development, validation and deployment, while also coordinating work across multidisciplinary teams. This experience has strengthened my ability to translate **open-ended research problems** into robust, **end-to-end computational solutions** under real-world constraints, complementing my previous academic research experience. In addition, I contribute to the design and implementation of **Physics-Informed Neural Networks (PINNs)**, bridging state-of-the-art scientific modeling with industrial **predictive maintenance and optimization**. I joined DEDUCE through the competitive Torres Quevedo 2024 program, which marked my transition from academia to applied research in the private sector.

Previously, **as a tenure track researcher at CEFCa**, I worked extensively on **astronomical image processing, PSF modeling and subtraction, source detection and low-surface-brightness analysis**, including crowded J-PLUS observations of IC1613 and deep ARRAKIS images. This work involved tools such as *AstroPhot* and *StarNet*, alongside the construction of PSF-photometric catalogs and the development of LSB processing workflows for Euclid (*Dimauro et al. in prep.*). I participated in the publication of several studies that applied **ML techniques on J-PAS/J-PLUS data** (*del Pino et al. 2024, López-Sanjuan et al. 2025, Domínguez-Sánchez et al. 2026*).

Also, I explored the use of ML algorithms to characterize **galactic sizes for Euclid** in a fully-automated way (**Vega-Ferrero et al. 2025**). I investigated **high-z galaxy morphologies in JWST/CEERS** observations, using a novel self-supervised ML approach (**Vega-Ferrero et al. 2024**). I created and publicly shared the **largest morphological catalog to date**, comprising over **27 million DES galaxies** (**Vega-Ferrero et al. 2021b**). I have contributed to several studies on galaxy morphology (*Siudek et al. 2026, Huertas-Company et al. 2026, Huertas-Company et al. 2024, Kartaltepe et al. 2023*), galaxy evolution (*Domínguez-Sánchez et al. 2025, Costantin et al. 2023a*), tidal feature detection (*Domínguez-Sánchez et al. 2023*), high-z stellar bars (*Costantin et al. 2023b, Guo et al. 2023*), and ultra-high-z galaxies (*Zavala et al. 2023, Finkelstein et al. 2022*), many of which apply ML techniques to astrophysical data. I am currently participating in one **proposal for JWST Cycle 4 (PI Raymond Simons)** on a census of **galaxy kinematics and outflows to $z \sim 7$** . In Cosmology, I specialized in **indirect dark matter (DM) detection** based on the **GL effect produced by GCs**. I presented the **first estimate of the Hubble parameter using multiply-lensed, resolved SN images** (**Vega-Ferrero et al. 2018**). Our methodology has been used by *Kelly et al. 2023a,b*, significantly reducing the uncertainty in Hubble parameter estimation. I have developed and shared **GL models for the six HST Frontier Field (FF) clusters**, facilitating **high-z galaxy lensing probability** analyses (**Vega-Ferrero et al. 2019**). My research delves into **lensing efficiencies and radial arcs statistics** in simulated DM halos within the LCDM and SIDM models (**Vega-Ferrero et al. 2021a**). Other contributions include studies on DM halo triaxiality

(**Vega-Ferrero et al. 2017**), ML approaches for GC mass estimation (*de Andrés et al. 2022*), analyses of accretion shocks in GCs (*Baxter et al. 2021*), pioneering work on compact DM constraints (*Diego et al. 2018b*), and the concentration-mass relation in GCs (*Meneghetti et al. 2014, Merten et al. 2015*).

Since 2019, when I published my first ML paper in astrophysics, I have **co-authored 13 publications developing or applying ML methods to astrophysical and astronomical problems, including 4 as first author**. More broadly, I am the **first author of 8 publications** and the **third author of 4**, which have collectively received **over 475 citations**.

My expertise and versatility allow me to **adapt to new research areas and tackle innovative problems in astrophysics**. I have a **strong mobility record**, having worked at 7 institutions across 3 countries, and visited leading research centers worldwide. Currently, I am a member of the Euclid Consortium, J-PLUS and J-PAS surveys, the CEERS collaboration within JWST, The Three Hundred project, and the upcoming ARRAKIHS mission. **I have mentored students at all levels**. My research has been featured in **international conferences, seminars, press releases**, and various outreach and **mass media**.

C. Relevant merits

C.1. Publications as first author

1. **Automated galaxy sizes in Euclid images using the Segment Anything Model**
Vega-Ferrero, J.; Buitrago, F.; Fernández-Iglesias, J.; Raji, S.; Sahelices, B.; Domínguez Sánchez, H.
Astronomy & Astrophysics, Volume 693, id.A232, 11 pp. (2025)
2. **On the nature of disks at high redshift seen by JWST/CEERS with contrastive learning and cosmological simulations**
Vega-Ferrero, J.; Huertas-Company, M.; Costantin, L.; Pérez-González, P. G. et al.
The Astrophysical Journal, Volume 961, Issue 1, id.51, 23 pp. (2024)
3. **Event detection and reconstruction using Neural Networks in TES devices: a case study for Athena/X-IFU**
Vega-Ferrero, J.; Ceballos, M.; Cobo, B.; Carrera, F. J.; García, P.; Puyol Gruart, J.
Publications of the Astronomical Society of the Pacific, Volume 134, Number 1032 (2022)
4. **Pushing automated morphological classifications to their limits with the Dark Energy Survey**
Vega-Ferrero, J.; H. Domínguez Sánchez; M. Bernardi; M. Huertas-Company; et al.
Monthly Notices of the Royal Astronomical Society, Volume 506, Issue 2 (2021)
5. **Constraining the cross-section of dark matter with giant radial arcs in galaxy clusters**
Vega-Ferrero, J.; J. M. Dana; J. M. Diego; G. Yepes; Cui, W.; M. Meneghetti
Monthly Notices of the Royal Astronomical Society, Volume 500, Issue 1 (2021)
6. **Probability of magnification in the Hubble Frontier Fields clusters**
Vega-Ferrero, J.; Diego, J. M.; Bernstein, G. M.
Monthly Notices of the Royal Astronomical Society, Volume 486, Issue 4 (2019)
7. **The Hubble Constant from SN Refsdal**
Vega-Ferrero, J.; Diego, J. M.; Miranda, V.; Bernstein, G. M.
The Astrophysical Journal Letters, Volume 853, Issue 2, article id. L31, 5 pp. (2018)
8. **On the shape of dark matter halos from MultiDark Planck simulations**
Vega-Ferrero, J.; Yepes, G.; Gottlöber, S.
Monthly Notices of the Royal Astronomical Society, Volume 467, Issue 3 (2017)

C.2. Publications as co-author

1. **Euclid Quick Data Release (Q1) Exploring galaxy properties with a multi-modal foundation model** (Euclid Collaboration; Siudek, M. et al.)
Astronomy & Astrophysics, Volume 711, id.A13, 28 pp. (2026)
2. **Euclid Quick Data Release (Q1), A first look at the fraction of bars in massive galaxies at $z < 1$** (Euclid Collaboration; Huertas-Company, M. et al.)
Astronomy & Astrophysics, Volume 711, id.A11, 14 pp. (2026)
3. **J-PAS: A neural network approach to single stellar population characterisation** (Domínguez Sánchez, H. et al.)
Astronomy & Astrophysics, Volume 705, id.A219, 16 pp. (2026)
4. **J-PLUS: Understanding outlier white dwarfs in the third data release via dimensionality reduction** (López-Sanjuan, C. et al.)
Astronomy & Astrophysics, Volume 701, id.A273, 18 pp. (2025)
5. **Euclid: I. Overview of the Euclid mission** (Euclid Collaboration; Mellier, Y. et al.)
Astronomy & Astrophysics, Volume 697, id.A1, 94 pp. (2025)
6. **The Cosmic Evolution Early Release Science Survey (CEERS)** (Finkelstein, S. L. et al.)
The Astrophysical Journal Letters, Volume 983, Issue 1, id.L4, 29 pp. (2025)
7. **J-PLUS: Bayesian object classification with a strum of BANNJOS** (del Pino, A. et al.)
Astronomy & Astrophysics, Volume 691, id.A221, 27 pp. (2024)
8. **Galaxy Morphology from $z \sim 6$ through the eyes of JWST** (Huertas-Company, M. et al.)
Astronomy & Astrophysics, Volume 685, id.A48, 28 pp. (2024)
9. **A Milky Way-like barred spiral galaxy at a redshift of 3** (Costantin, L. et al.)
Nature, Volume 623, Issue 7987, p.499-501 (2023)
10. **Identification of tidal features in deep optical galaxy images with convolutional neural networks** (Domínguez Sánchez, H. et al.)
Monthly Notices of the Royal Astronomical Society, Volume 521, Issue 3 (2023)
11. **Expectations of the size evolution of massive galaxies at $3 \leq z \leq 6$ from the TNG50 simulation: the CEERS/JWST view** (Costantin, Luca; Pérez-González, Pablo G.; Vega-Ferrero, Jesús; et al.)
The Astrophysical Journal, Volume 946, Issue 2, id.71, 17 pp. (2023)
12. **First Look at $z > 1$ Bars in the Rest-frame Near-infrared with JWST Early CEERS Imaging** (Guo Yuchen et al.)
The Astrophysical Journal Letters, Volume 946, Issue 1, id.L15, 17 pp. (2023)
13. **CEERS Key Paper. III. The Diversity of Galaxy Structure and Morphology at $z = 3-9$ with JWST** (Kartaltepe et al.)
The Astrophysical Journal Letters, Volume 945, Issue 1, id.L10, 13 pp. (2023)
14. **Dusty Starbursts Masquerading as Ultra-high Redshift Galaxies in JWST CEERS Observations** (Zavala Jorge A. et al.)
The Astrophysical Journal Letters, Volume 943, Issue 2, id.L9, 14 pp. (2023)
15. **Lessons Learned from the Two Largest Galaxy Morphological Classification Catalogues built by Convolutional Neural Networks** (T. Y. Cheng, H. Domínguez-Sánchez, J. Vega-Ferrero; et al.)
Monthly Notices of the Royal Astronomical Society, Volume 518, Issue 2, pp.2794-2809
16. **A Long Time Ago in a Galaxy Far, Far Away: A Candidate $z \sim 14$ Galaxy in Early JWST CEERS Imaging** (Finkelstein, Steven L. et al.)
The Astrophysical Journal Letters, Volume 940, Issue 2, id.L55, 15 pp. (2022)
17. **A Deep Learning Approach to Infer Galaxy Cluster Masses from Planck Compton- y parameter maps** (de Andrés, D. et al.)
Nature Astronomy, Volume 6, p. 1325-1331 (2022)

- 18. Shocks in the Stacked Sunyaev-Zel'dovich Profiles of Clusters I: Analysis with the Three Hundred Simulations**
(E. J. Baxter; S. Adhikari; **J. Vega-Ferrero**; W. Cui; C. Chang; B. Jain; A. Knebe)
Monthly Notices of the Royal Astronomical Society, Volume 508, Issue 2 (2021)
- 19. The Three Hundred project: a large catalogue of theoretically modeled galaxy clusters for cosmological and astrophysical applications** (Cui, W. et al.)
Monthly Notices of the Royal Astronomical Society, Volume 480, Issue 3 (2018)
- 20. CLUMP-3D. Testing LCDM with galaxy cluster shapes** (Sereno, M. et al.)
The Astrophysical Journal Letters, Volume 860, Issue 1, article id. L4, 6 pp. (2018)
- 21. Dark Matter under the microscope: constraining compact dark matter with caustic crossing events** (Diego, J. M. et al.)
The Astrophysical Journal, Volume 857, Issue 1, article id. 25, 27 pp. (2018)
- 22. A free-form lensing model of A370 revealing stellar mass dominated BCGs, in Hubble Frontier Fields images** (Diego, J. M. et al.)
Monthly Notices of the Royal Astronomical Society, Volume 473, Issue 4 (2018)
- 23. The Detection and Statistics of Giant Arcs behind CLASH Clusters** (Xu, B. et al.)
The Astrophysical Journal, Volume 817, Issue 2, article id. 85, 25 pp. (2016)
- 24. CLASH: The Concentration-Mass Relation of Galaxy Clusters** (Merten, J. et al.)
The Astrophysical Journal, Volume 806, Issue 1, article id. 4, 26 pp. (2015)
- 25. The MUSIC of CLASH: Predictions on the Concentration-Mass Relation**
(Meneghetti, M.; Rasia, E.; **Vega, J.**; Merten, J.; Postman, M.; Yepes, G.; et al.)
The Astrophysical Journal, Volume 797, Issue 1, article id. 34, 21 pp. (2014)
- 26. CLASH: The Enhanced Lensing Efficiency of the Highly Elongated Merging Cluster MACS J0416.1-2403** (Zitrin, A. et al.)
The Astrophysical Journal Letters, Volume 762, Issue 2, article id. L30, 6 pp. (2013)

C.3. Conferences

Invited lectures

- 2023 ESA Science Faculty workshop on machine learning meets galaxy classifica**
"A journey through the classification of galaxy images with Deep Learning"

Oral contributions

- 2025 Euclid - From Q1 to DR1** (La Laguna, Tenerife, Spain)
Automated galaxy sizes in Euclid images using the Segment Anything Model
- 2024 XVI Reunión Científica de la SEA** (Granada, Spain)
Identifying stellar disk truncations in Euclid galaxy images using SAM
- 2024 CEERS collaboration meeting** (San Lorenzo de El Escorial, Spain)
Truncations as a new galaxy size indicator
- 2023 Euclid Spain Meeting 2024** (ICE, Barcelona, Spain)
Identifying stellar disk truncations in Euclid galaxy images using SAM
- 2023 IAP/CCA 2023 conference: Debating the potential of machine learning in a:**
Data-driven galaxy morphologies at $z > 3$ with contrastive learning and cosmological

- 2022 European Astronomical Society Annual Meeting** (Valencia, Spain)
"Self-supervised learning for unveiling high- z galaxy structure in the JWST era"
- 2021 IAP Colloquium: "Debating the potential of machine learning in astronomical surveys"** (Paris, France)
"Pushing automated classifications to their limits: the largest galaxy morphological catalog up to date"
- 2019 First Spanish meeting on galaxy clusters** (Madrid, Spain)
"The Hubble constant from SN Refsdal" and "Constraining the cross-section of dark matter with giant radial arcs in galaxy clusters"
- 2019 VII Meeting in fundamental Cosmology** (Madrid, Spain)
"Constraining the cross section of dark matter with giant radial arcs in galaxy clusters"
- 2019 Dark Energy Survey collaboration meeting at UPenn** (Philadelphia, PA, USA)
"Pushing automated morphological classifications to their limits with DES"
- 2018 XIII Reunión Científica de la SEA** (Salamanca, Spain)
"The Hubble constant from SN Refsdal"
- 2018 Dark Matter 2018: From the smallest to the largest scales** (Santander, Spain)
"Testing self-interacting dark matter models with gravitational lensing"
- 2017 Galaxy Clusters 2017: Theory, observations & future developments** (Santander, Spain)
"Free-form lens models of four Hubble Frontier Fields Clusters"
Member of the Local Organizing Committee (LOC)
- 2015 MultidarkLens Workshop** (Marseille, France)
"The distribution function of Einstein radii as cosmological probes"
- 2015 Meeting in fundamental Cosmology** (Santander, Spain)
"The distribution function of Einstein radii from MUSIC clusters"

Poster contributions

- 2023 IAP/CCA 2023 conference: Debating the potential of machine learning in a:**
Identifying stellar disk truncations in Euclid galaxy images using SAM
- 2022 First Science Results from JWST** (STScI, Baltimore, MD, USA)
"Are disks galaxies the dominant morphology of the $z > 3$ Universe"
- 2020 ADASS XXX: Astronomy meets Granada online**
Granada, Spain
"Athena/X-IFU pulse identification using Convolutional Neural Networks"
- 2014 Cosmology with Galaxy Clusters in the XXI century** (Madrid, Spain)
"Tension in the rings: The distribution function of Einstein radii"
Member of the Local Organizing Committee (LOC)
- 2013 Tracing Cosmic Evolution with Clusters of Galaxies** (Seston, Italy)
"Tension in the rings: The distribution function of Einstein radii"
- 2013 Cluster Lensing: Peering into the Past, Planning for the Future** (Baltimore, MA, USA)
"Tension in the rings: The distribution function of Einstein radii"

Attended

- 2026 ARRAKIHS Consortium Meeting**
Santander, Spain
- 2025 European Astronomical Society Annual Meeting**
Cork, Ireland
- 2025 XX J-PAS meeting**
Teruel, Spain
- 2024 3rd ARRAKIHS Mission Consortium Meeting**
Grindelwald, Switzerland
- 2016 Dark Matter 2016: From the smallest to the largest scales**
Santander, Spain
- 2012 XMM-Newton 2012 Science Workshop: Galaxy Clusters as Giant Cosmic Laboratories**
Madrid, Spain
- 2011 Fronteras de la Computación**
Benasque, Spain
- 2005 Sistema Solar y otras familias de planetas**
Madrid, Spain
- 2004 Planetas y sistemas planetarios en el Universo: presente y futuro**
Madrid, Spain

C.4. Research projects

- 2023 Gobierno de Aragón (€100K)**
- 2025** "J-PHOT: Fotometría de precisión para los grandes cartografiados multifiltro del OAJ" (CEFCA, Teruel, Spain)
Participation as **Tenure Track researcher**
- 2021 MCIU, European Union-NextGenerationEU (€1M)**
- 2024** "J-CAVA: Catálogos de Alto Valor Añadido para los cartografiados multifiltro del OAJ" (CEFCA, Teruel, Spain)
Participation as **Tenure Track researcher**
- 2024 2023 Call for Faculty Research Proposals. ESA Science Directorate (€19K)**
Machine learning techniques applied to Galaxy Clusters: A membership determination tool based on the ESA Science Archive
Participating entities: ESA, UVa, CEFCA, CAB and IFCA
- 2021 Ministerio de Ciencia e Innovación (€99K)**
- 2024** Galactic Edges and Euclid in the Low Surface Brightness Era (GEELSBE); P.I.: F. Buitrago (UVa, Valladolid, Spain)
Participation as **post-doc researcher**
- 2020 EU Horizon2020 (€187K)**
- 2024** AHEAD2020, Integrated Activities for the High Energy Astrophysics Domain; Coordinator CSIC: M. T. Ceballos
Participation as **post-doc researcher**

- 2023 2022 Call for Faculty Research Proposals. ESA Science Directorate (€25K)**
Machine learning techniques applied to Galaxy Clusters: A membership determination tool based on the ESA Science Archive
Participating entities: ESA, UVa, CEFCa, CAB and IFCA
- 2019 MICIU, EU-FEDER and Agencia Canaria de Investigación, Innovación y Sociedad de la Información (€109K and €57K)**
"Evolución de galaxias con inteligencia artificial" (PC2018-100852-A-I00) and "Deep-ASTRONOMY" (ProID2020010057); P.I.: M. Huertas Company (IAC, La Laguna, Spain)
Participation as **post-doc researcher**
- 2019 National Science Foundation (NSF, USA) (\$200K)**
Deep Learning for Galaxy Morphology in the Big Data Era; P.I.: M. Bernardi (UPenn, Philadelphia, PA, USA)
Participation as **main post-doc researcher**
- 2015 Spanish government (National Plan: AYA) (€83K)**
2018 Lighting the Dark. Properties of dark matter through its impact on cosmological structures; P.I.: J. M. Diego (IFCA, Santander, Spain)
Participation as **post-doc researcher**
- 2016 HST/STScI (USA) (\$99.5K)**
2017 Free-Form Models for the Six Hubble Frontier Fields Clusters; P.I.: G. Bernstein (UPenn, Philadelphia, PA, USA) and J. M. Diego (IFCA, Santander, Spain)
Participation as **Frontier Fields postdoc**

C.5. Grants and fundings

- 2011 International Grant awarded from "Ministerio de Ciencia e Innovación 2015 (MICINN)"** to complete a PhD program at Universidad Autónoma de Madrid on Numerical Cosmology (Madrid, Spain)
- 2014 "Ayuda para estancias breves** en otros Centros de I+D del Subprograma de Formación de Personal Investigador del Ministerio de Ciencia e Innovación"
2-month visit (€3000) to collaborate with **Prof. David Valls-Gabaud** Observatoire de Paris (Paris, France).
- 2012 HPC-Europa2 Transnational Access programme**
5-week visit to collaborate with **Dr. Massimo Meneghetti** Observatory of Bologna (Bologna, Italy)
- 2012 "Ayuda para estancias breves** en otros Centros de I+D del Subprograma de Formación de Personal Investigador del Ministerio de Ciencia e Innovación"
4-month visit (€6600) to collaborate with **Prof. David Valls-Gabaud** Observatoire de Paris (Paris, France).

C.6. Surveys and collaborations

- 2024** Member of the [J-PLUS](#) and [J-PAS](#)
- 2023** Member of the **Structure Analysis Work Package** for [ARRAKIS](#) (ESA mission)
- 2021** Member of the [Euclid Consortium](#) (ESA mission)
- 2021** Member of the [Cosmic Evolution Early Release Science Survey \(CEERS\)](#)
- 2019** Member of the [Dark Energy Survey \(DES\)](#)
- 2017** Member of [The Three Hundred project](#)
- 2016** Member of the [Frontier Fields Lens Modeling v.4](#)

C.7. Contracts, technological or transfer merits

C.7.1. Value Added Catalogs (VAC)

- 2025 [IC 1613: PSF-photometry in twelve J-PLUS passbands](#)
- 2025 [Euclid galactic sizes](#)
- 2024 [J-PLUS BANNJOS object classifications](#)
- 2022 [JWST-CEERS Mock Galaxy Imaging](#)
- 2021 [DES Morphological Catalog](#)
- 2018 [Hubble Frontier Fields Lens models](#)

C.7.2. Seminars, meetings and workshops

- 2024 **Conference for “Máster Universitario en Inteligencia Artificial”** (UNIR)
“Avances y desafíos en el uso del aprendizaje automático en astrofísica”
- 2023 **CEFCA Seminar** (Teruel, Spain)
Seminar on "On the nature of disks at high redshift seen by JWST/CEERS with conti
- 2022 **XIII Día de Nuestra Ciencia 2022** (La Laguna, Spain)
Talk on "Unveiling galaxy structure at $z>3$ with JWST and self-supervised learning"
- 2022 **IAC Dark Energy Survey Data Workshop 2022** (La Laguna, Spain)
<https://research.iac.es/congreso/DESdataworkshop2022/>
Member of the Scientific Organizing Committee (SOC)
- 2021 **Instituto de Astrofísica de Canarias** (La Laguna, Spain)
Open Astronomy Seminar on "Pushing automated classifications to their limits: the
- 2021 **Instituto de Ciencias del Espacio** (Barcelona, Spain)
ICE Pizza Seminar on "Pushing automated classifications to their limits: the largest
- 2021 **Institut d'Investigació en Intel·ligència Artificial** (Barcelona, Spain)
Seminar on "TES X-ray pulse identification using Neural Networks"
- 2020 **The "Three Hundred" collaboration meeting** (Madrid, Spain)
Talk on "Cross section of dark matter with giant radial arcs in galaxy clusters"
- 2019 **Santa Cruz Galaxy Workshop** (Santa Cruz, CA, USA)
- 2017 **Crystal clear clusters** (Madrid, Spain)
Talk on "Gravitational lensing effect on The Three Hundred simulations"
- 2014 **nIFTy Cosmology Workshop** (Madrid, Spain)
- 2013 **CLUES Workshop 2013** (Madrid, Spain)
- 2012 **CLUES Workshop 2012** (Lyon, France)
Talk on "Density profiles of radiative simulated clusters"
- 2011 **EUCLID Mission Meeting 2011** (Bologna, Italy)
- 2011 Workshop on **Constrained Simulations of Galaxy and Large-Scale Structure Formation** (Sussex, UK)
Talk on "How many Gravitational lenses are there on the sky?"

C.7.3. Outreach and mass media

Press releases

- 2025** [AI and the size of galaxies](#) based on *Vega-Ferrero et al. 2025*
- 2023** [The most distant galaxy similar to the Milky Way](#) based on *Costantin et al. 2025*
- 2021** [27 million Galaxy morphologies](#) based on *Vega-Ferrero et al. 2021*

Mass media

- 2025** [La Ventana de Aragón](#)
- 2021** [RTVE](#), [Nvidia Developer](#), [La Nueva España](#), [Telemadrid](#), [The Register](#), [Europa Press](#)

C.8. Organization of events

- 2026** SOC for the ESA Space Science Faculty Workshop “[Machine Learning meets Galaxy Classification 2](#)” (Madrid, Spain)
- 2024** SOC and co-chair for the [SEA 2024](#) Symposia on “Deciphering the Low Surface Brightness Universe” (Granada, Spain)
- 2023** SOC [IAC Dark Energy Survey Workshop](#) (2022, La Laguna, Spain)
- 2021** LOC [Cosmology with Galaxy Clusters in the XXI century](#) (2014, Madrid, Spain)

C.9. Visits and short stays in research Institutions

Mar 2025 Invited visitor at UVa

5-days stay at the "Departamento de Física Teórica, Atómica y Óptica" (Valladolid, Spain)

2024 Invited visitor at IFCA

2025 Several stays for a total of 105 days (Santander, Spain)

Dec 2022 Invited visitor at Flatiron Institute (Simons Foundation)

8-days stay at the Center for Computational Astrophysics (New York, NY, USA)

Sep 2019 Invited visitor at Institute of Space Sciences (ICE-CSIC)

12 2019 97-days stay at the Department of Cosmology and Fundamental Physics (Barcelona, Spain)

Jul 2019 Invited visitor at University of California in Santa Cruz (UCSC)

Aug 2019 32-days stay at the Department of Astronomy and Astrophysics (Santa Cruz, CA, USA)

Aug 2017 Visiting scholar at University of Pennsylvania

Dec 2018 496-days stay at the Physics and Astronomy Department (Philadelphia, PA, USA)

C.10. Mentoring and teaching

2024 Co-supervision of Joyo Smit

Physics degree student at Saint Mary's University (SMU, Canada)

Project: self-supervised learning algorithms for the identification of tidal features in galaxy images of the Hyper Suprime-Cam (HSC) survey

Co-supervisors: Ivana Damjanov (SMU, Canada), H. Domínguez (IFCA, Spain)

- 2024 Co-supervision of TFG of Arturo Cano Moraleda (8.6/10 grade)**
Physics degree student at Universidad de Valladolid (UVa, Spain)
Co-supervisors: F. Buitrago (UVa, Spain)
- 2023 Co-supervision of "Prácticas de Empresa" of Arturo Cano Moraleda**
Physics degree student at Universidad de Valladolid (UVa, Spain)
- 2023 Co-supervision of TFM of Carlos Álvarez Fernández (9.5/10 grade)**
MSc student at Universidad Complutense de Madrid (UCM, Spain)
Project: "Discovering galaxy formation channels with unsupervised learning"
Co-supervisors: H. Domínguez (IFCA, Spain)
- 2019 Co-supervision of Alberto Manjón García**
Former PhD student at Instituto de Física de Cantabria (IFCA, Spain)
Project: Identifying galaxy-galaxy gravitational lenses using CNNs
Co-supervisors (IFCA, Spain): D. Herranz, J. M. Diego and H. Domínguez
- 2011 Assistant professor** for the Laboratory of Experimental Techniques in Physics
2012 within the Grade in Environmental Sciences
Faculty of Sciences, Universidad Autónoma de Madrid (Spain)